H Implementation of Safety Filters

All four filters were evaluated using their official, publicly available implementations on an NVIDIA H100 GPU server. Below, we describe the source and scoring method for each.

SD safety checker [33] was loaded from the HuggingFace `diffusers` library (`CompVis/stable-diffusion-safety-checker`) and invoked through `StableDiffusionSafetyChecker.forward()` with `CLIPImageProcessor` preprocessing, matching the reference diffusion pipeline. The continuous score is the maximum adjusted cosine similarity across 17 NSFW concept embeddings, and an image is flagged whenever any score exceeds 0.

Falconsai [21] was loaded from the official HuggingFace model card (`Falconsai/nsfw_image_detection`) via `AutoModelForImageClassification`. The continuous score is $P(\text{NSFW})$ from the softmax output of the two-class classifier, with the default decision threshold of $\tau = 0.5$.

Marqo NSFW-384 [20] was loaded from `Marqo/nsfw-image-detection-384` via the `timm` library, using its built-in data configuration and preprocessing transform. The continuous score is $P(\texttt{NSFW})$ from the softmax output, and images are flagged at the default threshold $\tau = 0.5$.

LlavaGuard v1.2-7B [23] was loaded from `AIML-TUDA/LlavaGuard-v1.2-7B-OV-hf` and prompted to output a structured JSON verdict (`"rating": "Safe"|"Unsafe"`). We evaluated two policies: the default LlavaGuard policy and a PSC-specific policy targeting political-figure impersonation and disinformation via visual narrative.